An Infinite Dimensional Version of Sard's Theorem

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Goal

- Generalize Sard's Theorem to infinite dimensional Banach manifolds and use it to tackle problems

Outline

① Sard's Theorem

② Problems involving infinite dimensional manifolds

③ Banach Manifolds

④ Smale's generalization

⑤ Some consequences

Sard's Theorem ① (1942)

Let M_1, M_2 be C^k-manifolds of dimensions n_1, n_2, respectively. Let $f: M_1 \to M_2$ be a C^k-map. If

$$k > \max(n_1 - n_2, 0),$$

> then the critical values of f have measure 0 in M_2.

We take our manifolds to be second countable. Thus, it suffices to show Sard's locally:

> Let f be a C^k-map of a neighborhood of D^{n_1} to $\mathbb{R}^{n_2}$. If $k > \max(n_1 - n_2, 0)$, then the critical values of f have measure 0.

$n_1 < n_2$

Indeed, it suffices to show the following lemma:

Lemma

> Let $f: U \to \mathbb{R}^n$ be a C^1-map, where $U \subseteq \mathbb{R}^n$ is open and $m > n$. Then, $f(U)$ has measure 0.

Proof

Define $F: U \times \mathbb{R}^{m-n} \to \mathbb{R}^m$ by $F(x,y) = f(x)$. Moreover, recall that $U \times \{0\}$ has measure 0 in $U \times \mathbb{R}^{m-n}$. Let $U_i = (U \times \{0\}) \cap B_i(0)$, so $U = \bigcup_i U_i$. Since f is C^1, so is F. Moreover, $U_i \subseteq \overline{B_i(0)}$, a compact set, so there is some $\lambda_i > 0$ such that

$$|F(x,y) - F(x',y')| \leq \lambda_i |(x,y) - (x',y')| \quad \text{on } U_i.$$

Given $\varepsilon > 0$, there exist solids S_j covering $U \times \{0\}$ such that

$$\sum_j \text{vol}(S_j) < \varepsilon / \lambda_i^m$$

Then,

$$\sum_j \text{vol}(F(S_j)) < \varepsilon$$

and the $F(S_j)$ cover $F(U \times \{0\}) = f(U)$. ■

$\Omega_1 = \Omega_2$

It suffices to show the following lemma:

Lemma

Let $f : D^n \to \mathbb{R}^n$ be defined and C^1 on a neighborhood of D^n. Then, the set of critical values of f has measure 0 in $\mathbb{R}^n$.

Proof

Since f is C^1 and D^n is compact, there is a constant $a > 0$ such that for $x, y \in D^n$

$$|f(y) - f(x)| < a|y - x| \tag{1}$$

Moreover, there is a function $b(\varepsilon)$ such that $b(\varepsilon) \to 0$ as $\varepsilon \to 0$ and

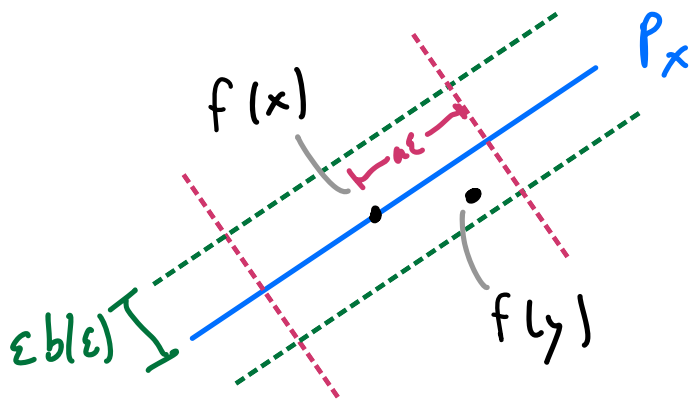
$$|f(y) - f(x) - df_x(y - x)| \leq b(|y - x|)|y - x| \tag{2}$$

Suppose x is a critical point of f. Then, $df_x(\mathbb{R}^n)$ is contained in some plane of dimension $n-1$. Let P_x be the translation of this plane by $f(x)$.

Suppose $|y - x| < \varepsilon$. Then,

$|f(y) - f(x)| < a\varepsilon$ and

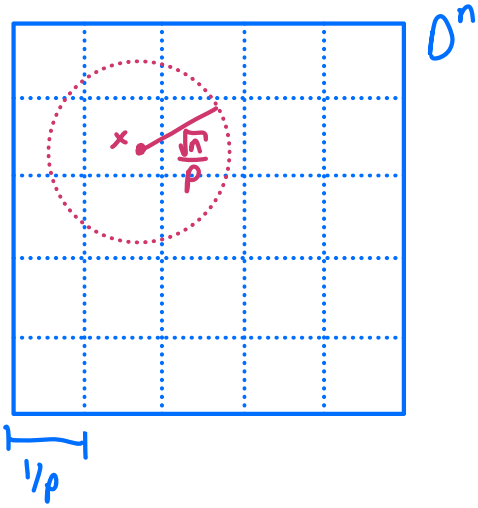
$d(f(y), P_x) < \varepsilon b(\varepsilon)$.



Thus, $f(y)$ lies in a rectangular prism of volume

$$\leq 2^n a^{n-1} \varepsilon^n b(\varepsilon).$$

Now, divide 0^n into p^n cubes of edge $1/p$. Observe that any cube containing a critical point x is contained in a ball of radius $\sqrt{n}/p$ centered at x. By our work above, the image of this ball has volume at most $2^n a^{n-1} (\sqrt{n}/p)^n b(\sqrt{n}/p)$. The set of critical values of f is contained in the image of the cubes containing at least one critical point. There are at most p^n such cubes, so the total volume is at most



$$2^n a^{n-1} n^{n/2} b(\sqrt{n}/p).$$

Taking $p \to \infty$, this expression $\to 0$. ∎

$n_1 > n_2$

Why does the same proof not work? Repeating our steps, we get a total volume of

$$2^{n_2} a^{n_2 - 1} n_1^{n_2/2} p^{n_1 - n_2} b(\sqrt{n_1}/p),$$

which for $n_1 > n_2$ does not vanish in general as $p \to \infty$.

To obtain a better estimate, we need the following results:

Lemma (Morse, 1939)

Let $A \subseteq \mathbb{R}^n$ and $k \geq 0$ an integer. Then, $A \subseteq \bigcup_{i=0}^{\infty} A_i$, where A_0 is countable and A_i for $i \geq 1$ satisfy that:
For any C^q function f defined on a neighborhood of A such that every point of A is a critical point of f, there exist functions $b_i(\varepsilon)$ such that $b_i(\varepsilon) \to 0$ as $\varepsilon \to 0$ and

$$|f(x) - f(y)| < b_i(|x-y|)\,|x-y|^k$$

Idea: If all derivatives of f vanish, this follows from Taylor expansion of f. If some partial derivative does not vanish at p, implicit function theorem gives a neighborhood N_p such that $N_p \cap A$ lies in a lower-dimensional submanifold. Then, do double induction on q and n.

Lemma

Let $f : D^{n_1} \to \mathbb{R}^{n_2}$ be defined and C^k on a neighborhood of D^{n_1}. Let A be the set of points where f has rank r. If

$$k \geq (n_1 - r)/(n_2 - r),$$

then $f(A)$ has measure 0.

Proof of Sard's

If $k > n_1 - n_2$, since $(n_2 - r - 1)(n_1 - n_2) \geq 0$ for $0 \leq r < n_2$,

$$(n_1 - n_2 + 1)(n_2 - r) \geq n_1 - r,$$

$$k \geq n_1 - n_2 + 1 \geq \frac{n_1 - r}{n_2 - r}.$$

Thus, if A is the set of critical values of f, $A = \bigcup_{r=0}^{n-1} f(A_r)$, where A_r is the set of points where f has rank r, and we can apply the lemma to each A_r to obtain that A has measure 0. ●

Motivation for Infinite Dimensions

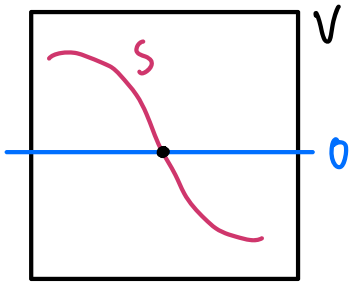
Even in the context of finite dimensional manifolds we find the need to handle infinite dimensional spaces.

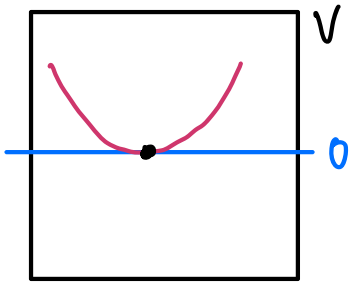
Let M be a finite dimensional manifold and $V \xrightarrow{\pi} M$ a vector bundle over M. We want to show that a generic section is transversal to the zero section.

The space of smooth sections $C^\infty(M, V)$ is infinite dimensional.

We have a map $C^\infty(M,V) \times M \xrightarrow{S} C^\infty(M,V) \times V$ defined by $S(s,x) = (s, s(x))$. Let $E = S^{-1}(C^\infty(M,V) \times \{0\})$.

Then, let $p : E \to C^\infty(M,V)$ be given by projection on the first coordinate. We can show that regular values of p correspond to sections transversal to the zero-section.





Thus, Sard's theorem would tell us that generic sections are transversal to the zero-section.

Say we want to show existence of solutions to ODEs:

$$\dot{x} = f(x), \quad x(0) = x_0.$$

Consider the map

$$C^1([0,\varepsilon], \mathbb{R}^n) \to C^0([0,\varepsilon], \mathbb{R}^n) \times \mathbb{R}^n$$
$$x \mapsto (\dot{x} - f(x), x_0).$$

If we could show this map is a diffeomorphism, then we would have the desired result. So, we also want some sort of Inverse Function Theorem in the infinite dimensional setting.

Banach Manifolds

Based on Lang's Intro to Diff. Geo.

A Banach space is a complete normed vector space. Given two Banach spaces E, F, we denote the space of bounded linear maps $E \to F$ by $L(E,F)$. Given $A: E \to F$, define

$$|A| = \sup\{K \mid |Ax| \leq K|x| \ \forall x \in E\}.$$

This norm makes $L(E,F)$ into a Banach space.

Proposition (Splitting)

> If E is a Banach space, and F_1, F_2 are two closed complementary ($F_1 + F_2 = E$, $F_1 \cap F_2 = \{0\}$) subspaces, then E is isomorphic to $F_1 \oplus F_2$.
>
> If $\dim F < \infty$ or F is closed and $\dim(E/F) < \infty$, then F always "splits" in this way.

Let E, F be Banach spaces and $U \subseteq E$ be open. Let $f: U \to F$ be a continuous map. We say f is differentiable at $x_0 \in U$ if there exists $\lambda \in L(E,F)$ such that if

$$f(x_0 + y) = f(x_0) + \lambda y + \varphi(y)$$

for small y, then $|\varphi(y)| \leq |y|\psi(y)$, where $\psi(y) \to 0$ as $|y| \to 0$. Then, we call λ the derivative of f at x_0. We denote it by $df_{x_0} \in L(E,F)$. If f is differentiable at every point of U, then

$$df : U \to L(E,F).$$

If df is continuous, we say f is of class C^1. Inductively define maps of class C^p. We say f is of class C^∞ if it is of class C^p for all $p \geq 1$.

We will need the following version of the mean value theorem:

Proposition

Let E, F be Banach spaces, $U \subseteq E$ open, and $x, z \in U$ distinct such that $x + t(z-x) \in U$ for all $t \in [0,1]$. Let $f: U \to F$ be of class C^1. Then,

$$|f(z) - f(x)| \leq |z - x| \sup_{\xi = x + t(z-x)} |df_\xi|$$

In this context, we can prove the Inverse and Implicit Function Theorems. First, we need the following lemma:

Lemma (Contraction)

> Let (M,d) be a complete metric space, and let $f: M \to M$ be a mapping of M into itself. Assume that there is a constant K, $0 < K < 1$, such that, for any two points $x, y \in M$, we have
>
> $$d(f(x), f(y)) \leq K\, d(x,y).$$

Then, f has a unique fixed point.

Proof

Existence: Let $x_0 \in M$ be arbitrary. We can show inductively that

$$d(f^j(x_0), x_0) \leq (1 + \dots + K^{j-1})\, d(f(x_0), x_0) \leq \frac{1}{1-K}\, d(f(x_0), x_0).$$

Thus, if $m > n > N$, we have

$$d(f^m(x_0), f^n(x_0)) \leq K^n\, d(f^{m-n}(x_0), x_0) \leq \frac{K^N}{1-K}\, d(f(x_0), x_0).$$

The right-hand side $\to 0$ as $N \to \infty$, so $f^n(x_0)$ is a Cauchy sequence. Since M is complete, it converges to some $x \in M$. For any ε, there is N such that $n > N$ implies $d(f^n(x_0), x) < \frac{\varepsilon}{1+K}$. Then,

$$d(x, f(x)) \leq d(f^{n+1}(x_0), x) + d(f^{n+1}(x_0), f(x)) < \varepsilon,$$

so taking $\varepsilon \to 0$, we see $f(x) = x$.

Uniqueness: Suppose y is another fixed point. Then,

$$d(x,y) = d(f^N(x), f^N(y)) \leq K^N\, d(x,y),$$

so taking $N \to \infty$, we see $x = y$. ∎

Inverse Function Theorem

Let E, F be Banach spaces, $U \subseteq E$ open, and $f: U \to F$ a map of class C^p, $p \geq 1$. Assume that for some $x_0 \in U$, $df_{x_0}: E \to F$ is an isomorphism. Then, there is an open neighborhood V of x_0 and $W \subseteq F$ open such that $f|_V: V \to W$ has an inverse $\varphi: W \to V$ of class C^p.

Proof

It suffices to show the result when $E = F$, $x_0 = 0$, $f(x_0) = 0$, and df_{x_0} is the identity. Let $g(x) = x - f(x)$. Then, $dg_{x_0} = 0$ and there exists $r > 0$ such that if $|x| < 2r$, then

$$|dg_x| < \frac{1}{2}.$$

From the mean value theorem, it follows $|g(x)| \leq \frac{1}{2}|x|$. Thus, g maps $\overline{B_r(0)} \to \overline{B_{\frac{r}{2}}(0)}$. We claim that for all $y \in \overline{B_{\frac{r}{2}}(0)}$, there exists a unique $x \in \overline{B_r(0)}$ such that $f(x) = y$. Consider $g_y(x) = y + g(x)$. If $|y| \leq \frac{r}{2}$, $|x| \leq r$, then

$$|g_y(x)| \leq |y| + |g(x)| \leq r,$$

so g is a map of the complete metric space $\overline{B_r(0)}$ into itself. Moreover,

$$|g_y(x_1) - g_y(x_2)| = |g(x_1) - g(x_2)| \leq \frac{1}{2}|x_1 - x_2|,$$

so g_y is a contraction map. By the contraction lemma, g_y

has a unique fixed point. A fixed point of g_y is exactly a solution of $f(x) = y$.

Hence, we obtain $\varphi : B_{\frac{r}{2}}(0) \to B_1(0)$ a local inverse of f.

Since

$$\begin{aligned} |x_1 - x_2| &\leq |f(x_1) - f(x_2)| + |g(x_1) - g(x_2)| \\ &\leq |f(x_1) - f(x_2)| + \tfrac{1}{2}|x_1 - x_2|, \end{aligned}$$

it follows $|x_1 - x_2| \leq 2|f(x_1) - f(x_2)|$. Thus, φ is continuous.

Moreover,

$$\begin{aligned} |\varphi(y_1) - \varphi(y_2) - df^{-1}_{x_2}(y_1 - y_2)| &= |x_1 - x_2 - df^{-1}_{x_2}|f(x_1) - f(x_2)|| \\ &\leq A\,|df_{x_2}(x_1 - x_2) - f(x_1) + f(x_2)|, \end{aligned}$$

which we can bound by differentiability of f. Thus, φ is differentiable and

$$d\varphi_y = df^{-1}_{\varphi(y)}, \quad y \in B_{\frac{r}{2}}(0).$$

Now, df is C^{p-1}, φ is continuous, and taking inverses is C^∞, so C^1. Continuing inductively, we deduce φ is C^p. ∎

This last bit of logic illustrates a concept known as bootstrapping:

Proposition (Bootstrapping)

If $f : U \to V$ is a homeomorphism of class C^p, $p \geq 1$ and its inverse is of class C^1, then its inverse is of class C^p.

Similarly, we can show the Implicit Function Theorem.

An atlas of class C^p on M is a collection of pairs (U_i, φ_i) (called charts) such that

1) Each $U_i \subseteq M$ open and $\bigcup U_i = X$.

2) Each φ_i is a bijection of U_i onto an open subset of some Banach space E_i, and for any i, j, $\varphi_i(U_i \cap U_j) \subseteq E_i$ open.

3) The map $\varphi_j \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ is of class C^p and has inverse of class C^p for each pair of indices i, j.

Given $U \subseteq M$ open and $\varphi : U \to U' \subseteq E$ open, we say (U, φ) is compatible with the atlas if each $\varphi_i \varphi^{-1}$ is of class C^p with inverse of class C^p.

Two atlases are compatible if each chart of one is compatible with the other. Compatibility defines an equivalence relation. An equivalence class of atlases of class C^p define a C^p-manifold structure on M.

If all the Banach spaces E_i in some atlas are equal to E, we say M is an E-manifold or that M is modeled on E.

Let $x \in M$. Consider triples (U, φ, v) where (U, φ) is a chart at x and $v \in E$. Say (U, φ, v) and (V, ψ, w) equivalent if

$$d(\psi \varphi^{-1})_{\varphi(x)}(v) = w.$$

An equivalence class of such triples is called a tangent vector at x. The set of all tangent vectors is the tangent space T_xM. Mapping $(U, \varphi, v) \mapsto v$ gives a bijection of T_xX and E, so that T_xM is a Banach space.

A map $f: M \to N$ is of class C^p if $\forall x \in M$ there is a chart (U, φ) at x, (V, ψ) at $f(x)$ with $f(U) \subseteq V$ such that $\psi \circ f \circ \varphi^{-1}$ is C^p.

Then, $df_x : T_xM \to T_{f(x)}N$ is the unique linear map that makes

$$\begin{array}{ccc} T_xM & \xrightarrow{\cong} & E \\ {\scriptstyle df_x}\downarrow & & \downarrow{\scriptstyle d(\psi f \varphi^{-1})_x} \\ T_{f(x)}N & \xrightarrow{\cong} & F \end{array}$$

commute.

Sard's Theorem for Banach Manifolds ④

In order to show the infinite dimensional analogue of Sard's theorem, we need to discuss Fredholm operators.

A map $A \in L(E, F)$ is Fredholm if

a) $\dim \ker A < \infty$

b) $A(E)$ is closed

c) $\dim \operatorname{coker} A < \infty$.

If A is Fredholm, its index is defined as the integer

$$\operatorname{ind} A := \dim \ker A - \dim \operatorname{coker} A.$$

Fact: The set $\mathrm{Fred}(E,F) \subseteq L(E,F)$ of Fredholm operators is open and $\mathrm{ind}: \mathrm{Fred}(E,F) \to \mathbb{Z}$ is continuous.

A Fredholm map between manifolds is a C^1 map $f: M \to N$ such that for each $x \in M$, $df_x : T_xM \to T_{f(x)}N$ is Fredholm. The index of f is defined as

$$\mathrm{ind}\, f := \mathrm{ind}\, df_x$$

for some $x \in M$. If M is connected, this does not depend on x.

Just as in the finite dimensional case, a point $x \in M$ is called regular if df_x is surjective and critical otherwise. The images under f of critical points are called critical values and all other points in N are called regular values.

Before stating and proving Smale's Sard's Theorem, we need the following results on Fredholm maps:

Lemma (Local Fredholm Maps)

Let $f: M \to N$ be a Fredholm map. Then, for any $x \in M$, there are Banach spaces B, K, C with $\dim K = \dim \ker df_x$, $\dim C = \dim \mathrm{coker}\, df_x$, and charts $\varphi: U \subseteq M \to B \oplus K$ and $\psi: V \subseteq N \to B \oplus C$ such that

$$(\psi \circ f \circ \varphi^{-1})(x,k) = (x, g(x,k))$$

for some $g: \varphi(U) \to C$.

Proof:

If we have charts $\tilde{\varphi}: \tilde{U} \to E$, $\tilde{\psi}: \tilde{V} \to F$, with $\tilde{\varphi}(x) = 0$, $\tilde{\psi}(f(x)) = 0$, then $d(\tilde{\psi} \circ f \circ \tilde{\varphi}^{-1})_0 = A : E \to F$ is Fredholm.

Let $K = \ker A$. Since $\dim K < \infty$, K splits, so there is some B_1 such that $E = B_1 \oplus K$. Let $B_2 = \operatorname{im} A$. Since $\dim F/B_2 < \infty$, B_2 splits, so it has a closed complement C such that $F = B_2 \oplus C$. Moreover, $A|_{B_1}$ is an isomorphism between B_1 and B_2, so without loss of generality, $B_1 = B_2 = B$ and $A|_{B_1}$ is the identity. Note that

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Write $(\tilde{\psi} \circ f \circ \tilde{\varphi}^{-1})(x, K) = (a(x,K), b(x,K))$ for $a: B \oplus K \to B$ and $b: B \oplus K \to C$. Consider the map $h: \tilde{\varphi}(\tilde{U}) \to B \oplus K$ given by

$$h(x, K) = (a(x,K), K).$$

Then,

$$dh_{(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is invertible, so the Inverse Function Theorem gives a local inverse g of h. Then,

$$(\tilde{\psi} \circ f \circ \tilde{\varphi}^{-1} \circ g)(x, K) = (x, g(x, K)),$$

as needed. ∎

Lemma (Locally Closed)

> A Fredholm map $f: M \to N$ is locally closed.

Proof

Choose charts as guaranteed by the previous lemma, so we may assume

$$f(x,K) = (x, g(x,K)).$$

Let $S \subseteq U \subseteq B \oplus K$ be closed and (x_i, c_i) be a sequence in $f(S)$ that converges to (x,c). Since S is closed, the x_i converge to some $x \in S$. The $c_i = g(x_i, y_i)$ for some $y_i \in K$. But K is closed, so the y_i converge to $y \in K$. Hence, the c_i converge to c in $f(S)$, so $f(S)$ is closed. ∎

Theorem (Smale, 1964)

> Let $f: M \to N$ be a C^k-Fredholm map with
>
> $$k > \max(\operatorname{ind} f, 0).$$
>
> Then, the regular values of f are a countable intersection of open dense sets. (= residual / comeager)

Proof

Since M is second countable, it suffices to prove this locally. Thus, $f|_U : U \to F$, where $U \subseteq E$ open. Since f is locally closed and the critical point set of f is closed, we may choose U so that the regular values of $f|_U$ are an open set.

Moreover, choose these charts so that f has the form guaranteed by our previous lemmas: $f(x,K) = (x, g|_{x,K})$.

Then,

$$df_{(x,K)} = \begin{pmatrix} 1 & 0 \\ * & \left(dg|_{\{x\}\times K}\right)_{(x,K)} \end{pmatrix}.$$

Now, $df_{(x,K)}$ is surjective if and only if $\left(dg|_{\{x\}\times K}\right)_{(x,K)}$ is surjective. But

$$\left(dg|_{\{x\}\times K}\right)_{(x,K)} : \{x\}\times K \longrightarrow C$$

is a map of finite dimensional Banach spaces. Moreover, if f is of class C^k, we can check g is of class C^k, with

$$k > \max(\operatorname{ind} f, 0) = \max(\dim K - \dim C, 0),$$

so Sard's theorem applies. Thus, the regular values at each slice $\{x\}\times K$ are dense, and thus the regular values of $f|_U$ are dense. ∎

⑤

Consequences

Corollary

If $\operatorname{ind} f < 0$, then $\operatorname{int} f(M)$ is empty.

Corollary

Under the conditions of the theorem, for "almost all" $y \in N$, $f^{-1}(y)$ is a submanifold of M of dimension $\operatorname{ind} f$ or is empty.

Theorem

Let $f: M \to N$ be a C^k-Fredholm map and $v: X \hookrightarrow N$ a C^1 embedding of a finite dimensional manifold X, with

$$k > \max(\operatorname{ind} f + \dim X, 0).$$

Then, there exists a C^1 approximation g' of g such that $f \pitchfork g'$.

Theorem

Let $f: M \to N$ be a Fredholm map transversal to the embedding $v: X \hookrightarrow N$, $\dim X < \infty$. Then, $f^{-1}(v(X))$ is a submanifold of M of dimension $\operatorname{ind} f + \dim W$.

We can use this to define a generalized degree mod 2 as

$$\gamma(f) = \text{cobordism class of } f^{-1}(y)$$
$$\text{for } y \text{ a regular point}$$
$$\text{of } f.$$